



The University of Georgia

ECSE 2920: Design Methodology

Deliverable 2

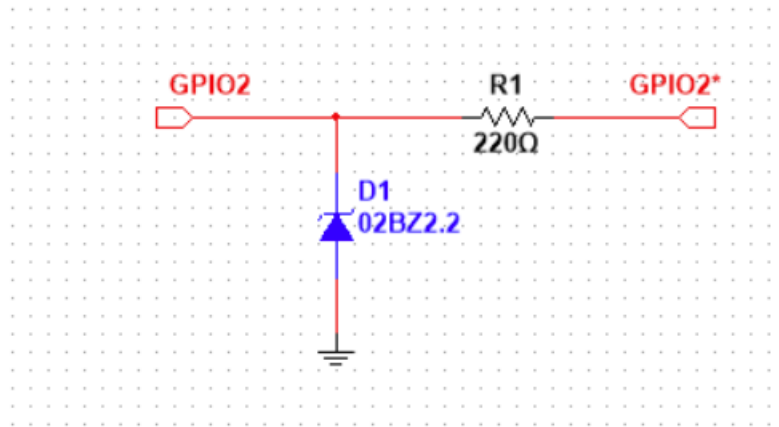
Group 16

01/15/2025

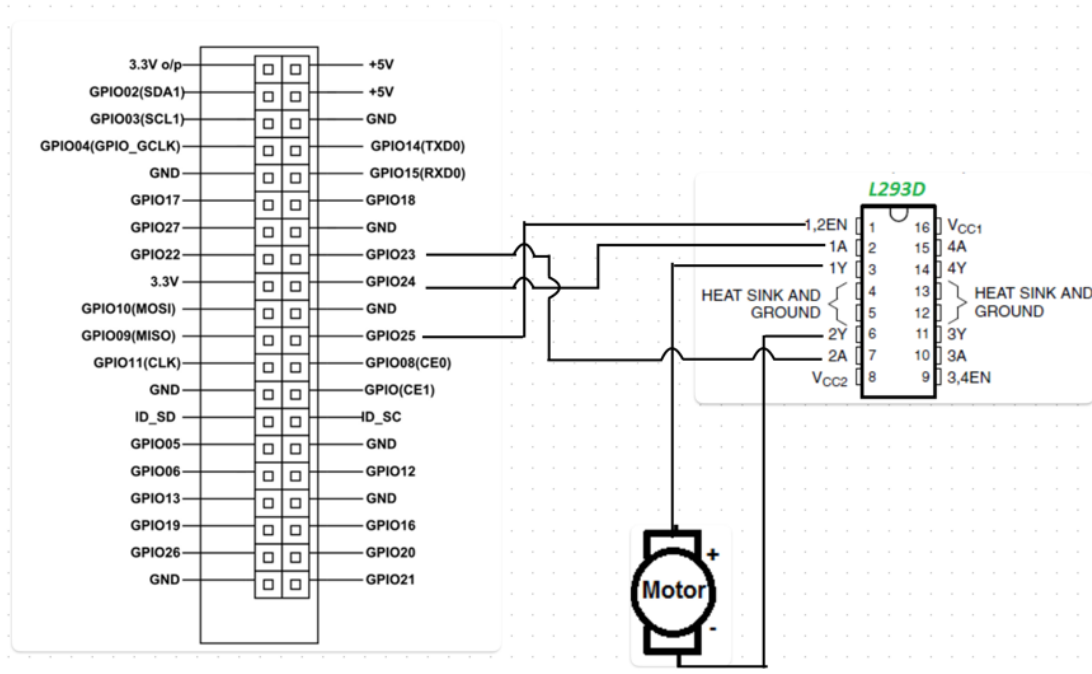
Part 1: Circuit Diagrams for the GPIO2 and H-Bridge

Requirements: The requirements for this part consist of understanding the GPIO2 on the PHP and how the H-Bridge works with the RP4 and DC motor connections. The purpose of the circuit diagram is to understand the functionality of the GPIO2 in the design and how the H-Bridge would affect the raspberry pi and the DC motors.

Picture(s)



The PHP acts as a secure way to connect devices to the raspberry pi. Connecting directly to the pi could cause various problems, and therefore the PHP is used as a layer of protection for the pi when connecting devices to the GPIO pins. The diagram we provided details the connection of GPIO2 on the raspberry pi to the GPIO2 pin on the PHP. The Zener diode in the diagram allows current to flow in both directions and is used as a stable voltage reference. It essentially allows current to flow from the pin on the PHP to the pin on the raspberry pi while adding a layer of security and stability which protects the pi from having too much power drawn.



The L293 H-Bridge serves as a mediator between the Raspberry Pi 4 and the DC motor, allowing the Pi to safely control the motor's operation. GPIO pins from the Raspberry Pi are connected to the input pins (2A, 1A, and 1,2 EN) on the L293, which determine the motor's direction and speed. The L293 outputs (1Y and 2Y) are connected to the DC motor terminals, providing the necessary voltage and current to drive the motor. This setup protects the Raspberry Pi from handling the motor's demands, while enabling precise control of motor operation through the Pi's GPIO pins.

Key Design Decision(s)

- **What did your team clarify about the design?**
We clarified how we all would have access to GitHub and how we would connect our pi to our GitHub repository.
- **What were the competing choices in the design?**
So far, we have not had any competing choices for the design as this week was mostly set up and downloading software.
- **Ultimately what did your team choose and why?**
We all agreed very well on what to do and how to complete our deliverables.

TEST... Test... test

- **What aspects of the design need to be tested?**
 - **Software?** In terms of software, the git commands were needed and tested for all team members to update the Group 16 GitHub repository for any changes.

- **Hardware?** In terms of hardware, the group members familiarized themselves with the use of the GPIO2 and H-Bridge parts to understand how those components will be used in our design.
- **Who is responsible for testing?**
 - **Tests ran?**
 - We did not test any components as we primarily created diagrams demonstrating the connections between components.
 - **Conclusions from testing?**
 - No conclusions were made as no testing was conducted.

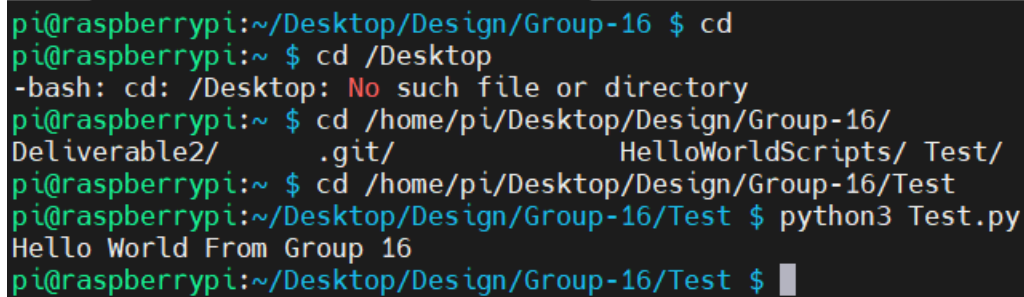
Summary/Part conclusion (make sure you address all parts of the requirements)

In conclusion, this section of the design emphasizes the importance of understanding both the functionality of the GPIO2 on the PHP and the interaction between the H-Bridge and the Raspberry Pi 4 in controlling the DC motor. The use of the PHP as a protective layer ensures the safe connection between external devices and the Raspberry Pi, preventing potential issues from directly connecting to the GPIO pins. The Zener diode further stabilizes the voltage reference, ensuring smooth and secure operation. Meanwhile, the L293 H-Bridge serves as an effective mediator, allowing precise control of the DC motor while shielding the Pi from high power demands.

Part 2: (Raspberry Pi 4 Setup)

Requirements: The requirements were setting up and using the Raspberry Pi 4 with python. The Raspberry Pi 4 should have a microSD card with Raspberry Pi OS installed, along with Python 3 and necessary libraries. Git must be configured with SSH keys for team members to collaborate effectively on the repository. Using the RP4, “Hello World” script was created and pushed it to GitHub.

Picture(s)



```
pi@raspberrypi:~/Desktop/Design/Group-16 $ cd
pi@raspberrypi:~ $ cd /Desktop
-bash: cd: /Desktop: No such file or directory
pi@raspberrypi:~ $ cd /home/pi/Desktop/Design/Group-16/
Deliverable2/      .git/              HelloWorldScripts/ Test/
pi@raspberrypi:~ $ cd /home/pi/Desktop/Design/Group-16/Test
pi@raspberrypi:~/Desktop/Design/Group-16/Test $ python3 Test.py
Hello World From Group 16
pi@raspberrypi:~/Desktop/Design/Group-16/Test $
```

Key Design Decision(s)

- **What did your team clarify about the design?**
The SD card needs to be in the pi to be able to boot and it is essential to have an ethernet connection for a smooth experience with the pi. Additionally, the Wi-Fi is difficult to connect to without an Ethernet cable, so it is beneficial to have one on standby.
- **What were the competing choices in the design?**
So far, we have not had any competing choices as we spent most of our time this week mostly working on setting up our pi for when we will need it in the future.
- **Ultimately what did your team choose and why?**
We did not have any big choices to make as we mostly just had to follow the setup instructions.

TEST... Test... test

- **What aspects of the design need to be tested?**
 - **Software?** We went over the setup for the raspberry pi and clarified the correct code setup for the RP4.
 - **Hardware?** In terms of hardware, we looked at the raspberry pi and understood the components of it which will be incorporated into the entire design.
- **Who is responsible for testing?**
 - Tests ran?

- The code required to create the message of “Hello World, from Group 16”.
- Conclusions from testing?
 - An ethernet cord had to be utilized for the Raspberry Pi 4 to connect to the WIFI so we could complete the push the GitHub. We successfully displayed the message and updated our GitHub repository.

Summary/Part conclusion (make sure you address all parts of the requirements)

In conclusion, this phase focused on setting up the Raspberry Pi 4 with Python, ensuring the microSD card had Raspberry Pi OS and necessary libraries, and configuring Git with SSH keys for team collaboration. The "Hello World" script was created and pushed to GitHub. The team clarified the need for the microSD card for booting and an Ethernet connection for a smoother experience.

Conclusion and Participation (REQUIRED)

1. Include a selfie/photo from your group meeting/zoom call.



2. **When did you meet?**
 - a. Met Monday in room 1409 at 1:30 pm
 - b. Met Tuesday in room 1459 at 11:00 am
3. **Who was present?**
 - a. Everyone was present on both days.
4. **Who was not present?**
 - a. No one did not show up.
5. **What were the main ideas discussed or major decisions (1-2 sentences/bullet points)**
 - a. Completions of the diagrams and documents
 - b. How to commit from a Raspberry Pi 4
 - c. Importance of an ethernet bridge